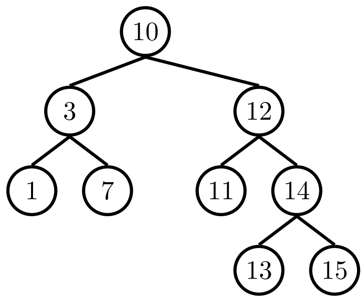
CS 61B — Traversals, Tries, Heaps

Spring 2021 — Discussion 9: March 15, 2020

# 1 Tree-versals




Write the pre-order, in-order, post-order, and level-order traversals of the above binary search tree.

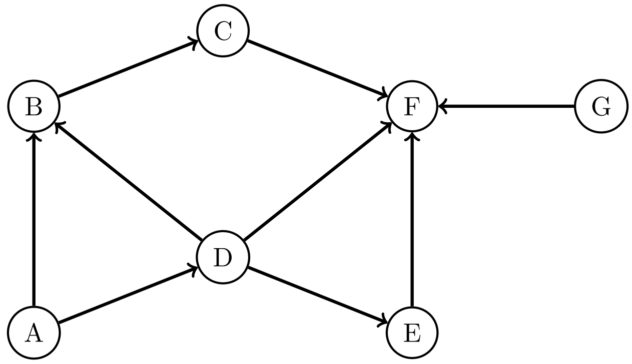
Pre-order: 10 3 1 7 12 11 14 13 15

In-order: 1 3 7 10 11 12 13 14 15

Post-order: 1 7 3 11 13 15 14 12 10

Level-order (BFS): 10 3 12 1 7 11 14 13 15

# 2 Graph Representations




(a) Write the graph above as an adjacency matrix, then as an adjacency list. What would be different if the graph were undirected instead?

0 1 0 1 0 0 0
0 0 1 0 0 0 0
0 0 0 0 0 1 0
0 1 0 0 1 1 0
0 0 0 0 0 1 0
0 0 0 0 0 0 0
0 0 0 0 0 1 0

A [B, D)
B [C]
C: [F]
D: [B, E, F]
E: [F]
F: [ ]
G: [F]

0 1 0 1 0 0 0
1 0 1 1 0 0 0
0 1 0 0 0 1 0
1 1 0 0 1 1 0
0 0 0 1 0 1 0
0 0 1 1 1 0 1
0 0 0 0 0 1 0

A: [B, D]
B: [A, C, D]
C: [B, F]
D: [A B E F]
E: [D F]
F: [C D E G]
G: [F]

(b) Write the order in which DFS pre-order graph traversal would visit nodes in the directed graph above, starting from vertex $A$. Break ties alphabetically. *Extra: Do the same for DFS post-order and BFS*

A B C F D E

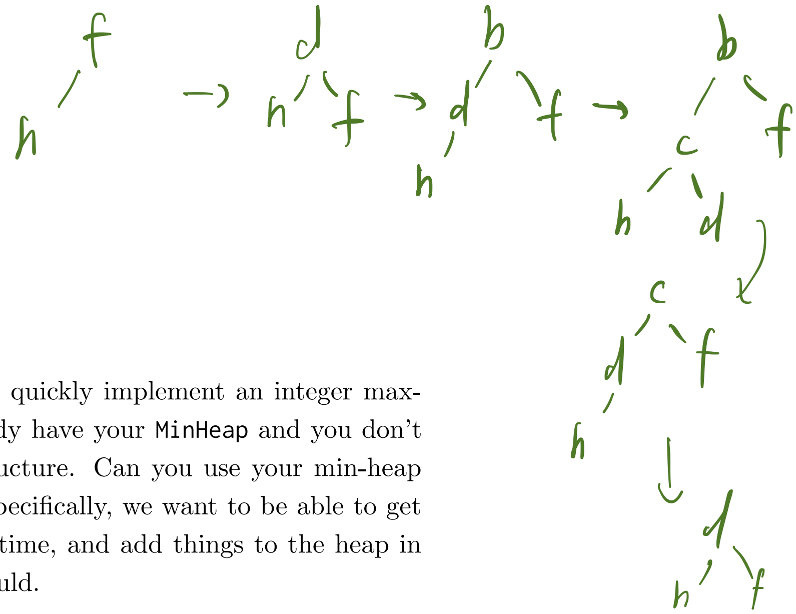
F C B E D A

A B D C E F

# 3 Heaps of Fun

(a) Assume that we have a binary min-heap (smallest value on top) data structure called `MinHeap` that has properly implemented `insert` and `removeMin` methods. Draw the heap and its corresponding array representation after each of the operations below:

```
Heap<Character> h = new MinHeap<>();
h.insert('f');
h.insert('h');
h.insert('d');
h.insert('b');
h.insert('c');
h.removeMin();
h.removeMin();
```



(b) Your friendly TA Tony challenges you to quickly implement an integer max-heap data structure. However, you already have your `MinHeap` and you don't feel like writing a whole second data structure. Can you use your min-heap to mimic the behavior of a max-heap? Specifically, we want to be able to get the largest item in the heap in constant time, and add things to the heap in $\Theta(\log n)$ time, as a normal max heap should.

*Hint*: Although you cannot alter them, you can still use methods from `MinHeap`.

Max-Heap.insert(A) = minHeap.insert(-A);

maxHeap.removeMax() = -minHeap.removeMin();